EDAB6808 · ASSIGNMENT 4

NLP of Bitcoin Tweets

Topic modelling and sentiment-volatility analysis on 4.69 million tweets

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Introduction: objective and dataset

Objective

Question 1 · Topic modelling

Discover the dominant themes in Bitcoin-related tweets and explain how those themes change over time.

Question 2 · Sentiment and volatility

Score sentiment polarity on the unlabelled tweets and test whether sentiment helps explain or predict Bitcoin volatility, beyond market variables alone.

Dataset

4.69M

tweets after a correct parse

13 → 7

columns available → used

12 months

6 Mar 2025 to 2 Mar 2026

201 days

actually carry tweets

Tweets: Bitcoin Tweets Dataset 2025-2026 (Kaggle). Text, date, user, followers, source, hashtags and retweet flag.

Prices: BTC-USD daily OHLC from Yahoo Finance, for the volatility target.

Three findings

2.4x

A default CSV read overstates the corpus. Real size is 4,689,305 tweets, not 11.2 million.

18.1%

Of the modelling sample is promotional template, occupying five of the ten topics found.

4 years

The dataset's timestamps are shifted. Content is from 2021-22, dates say 2025-26.

Sentiment did not improve volatility forecasting. The sentiment block fails its joint F-test at $p = 0.373$, and every model sits at or below 50% directional accuracy.

What does predict next-day volatility is market structure: yesterday's return ($p < 0.001$) and yesterday's trading range ($p = 0.007$). A six-variable linear model beat both ensembles.

The file would not parse

- Newlines inside quoted tweets split one tweet across several rows
- Quotes escaped with a backslash, not doubled, against RFC 4180
- Some tweets contain backslashes as ordinary text, so no single setting works
- PyArrow with explicit escape handling: 50 rows skipped out of 4.69 million, 0.001%

11,198,521

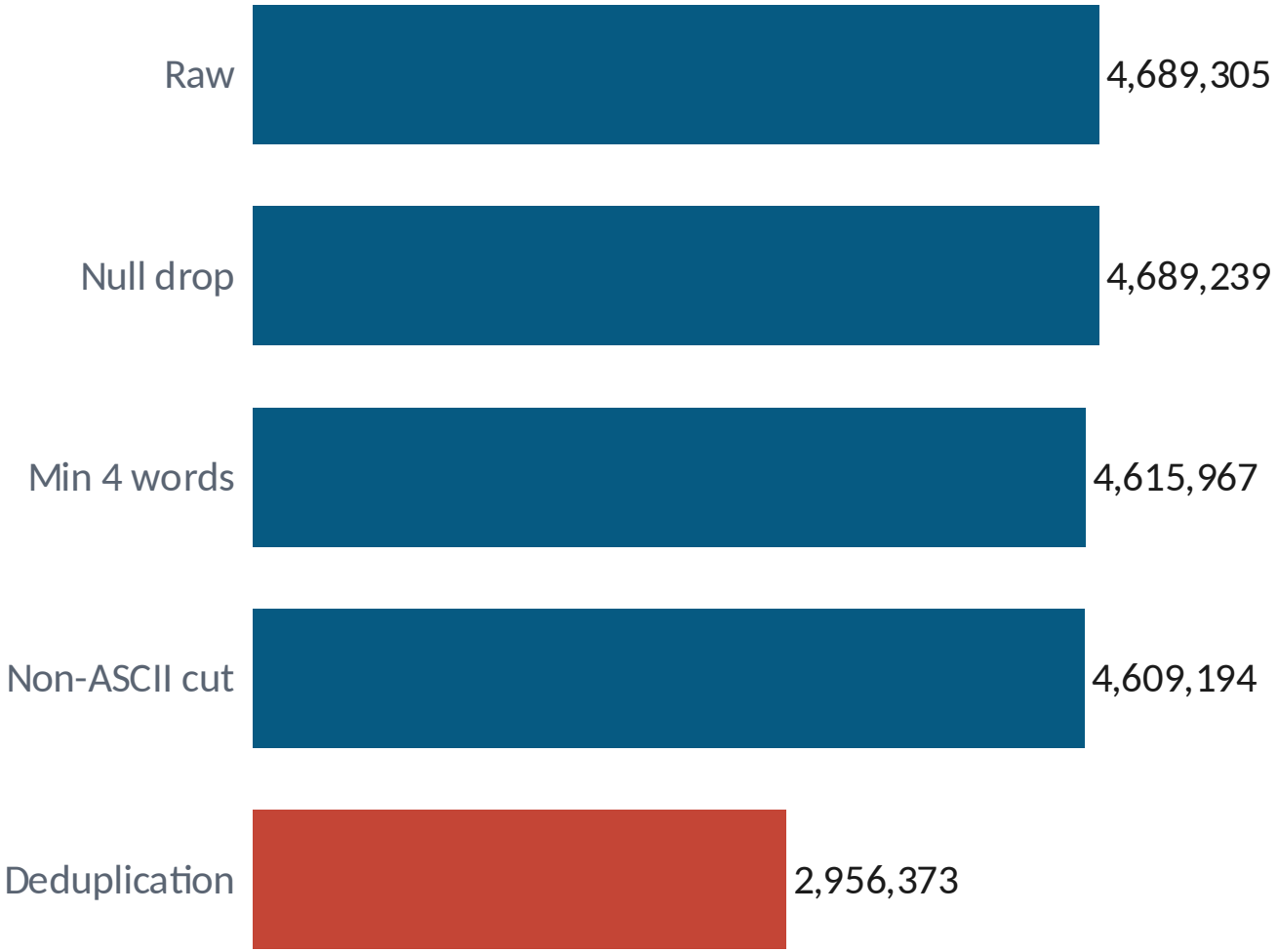
rows reported by a default read

4,689,305

actual tweets in the corpus

Every column first typed as String, including follower counts, because misaligned fields pushed tweet text into numeric columns. Correct types on the second read were the signal the parse was right.

37% of the corpus is duplicate



The filters cost 1.7% between them. Deduplication removes the other 35.3%.

- Raw duplicate count was only 2.5%
- Stripping URLs exposed the rest: same body, different tracking link
- 1,886 templates posted 100+ times
- They account for 823,596 rows

Catching the bots dedupe misses

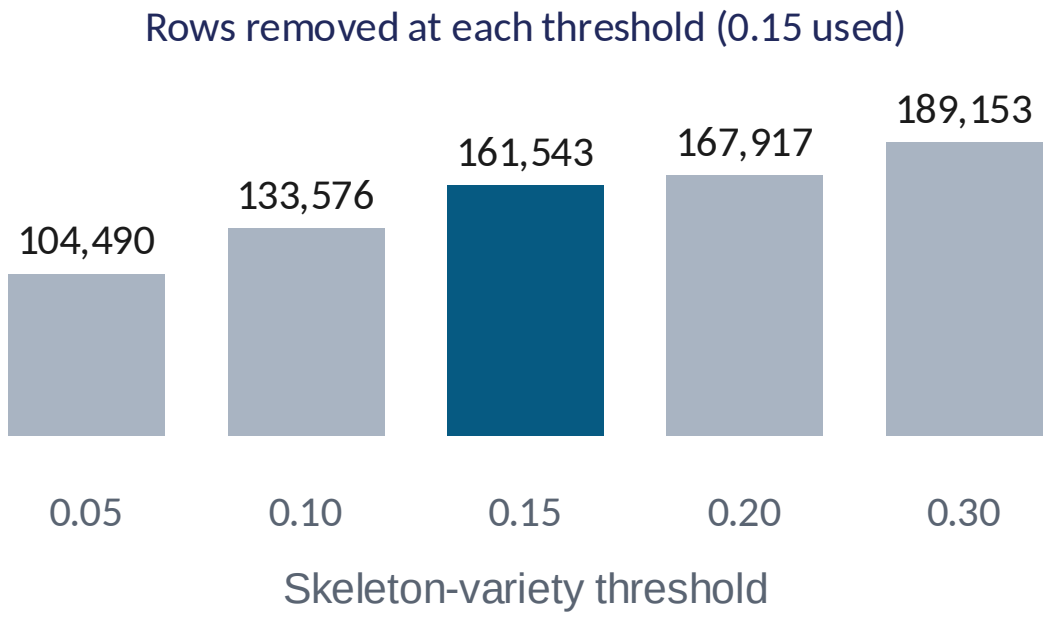
A ticker posting a new price every ten minutes is textually unique every time, so deduplication never sees it.

The method

Cut each tweet to its first five words, call that the skeleton, and score each account on how many distinct skeletons it produces.

Bitswings: 9,707 posts, 9 skeletons.
Bitcoin Whale Alert: 3,195 posts, 1 skeleton.

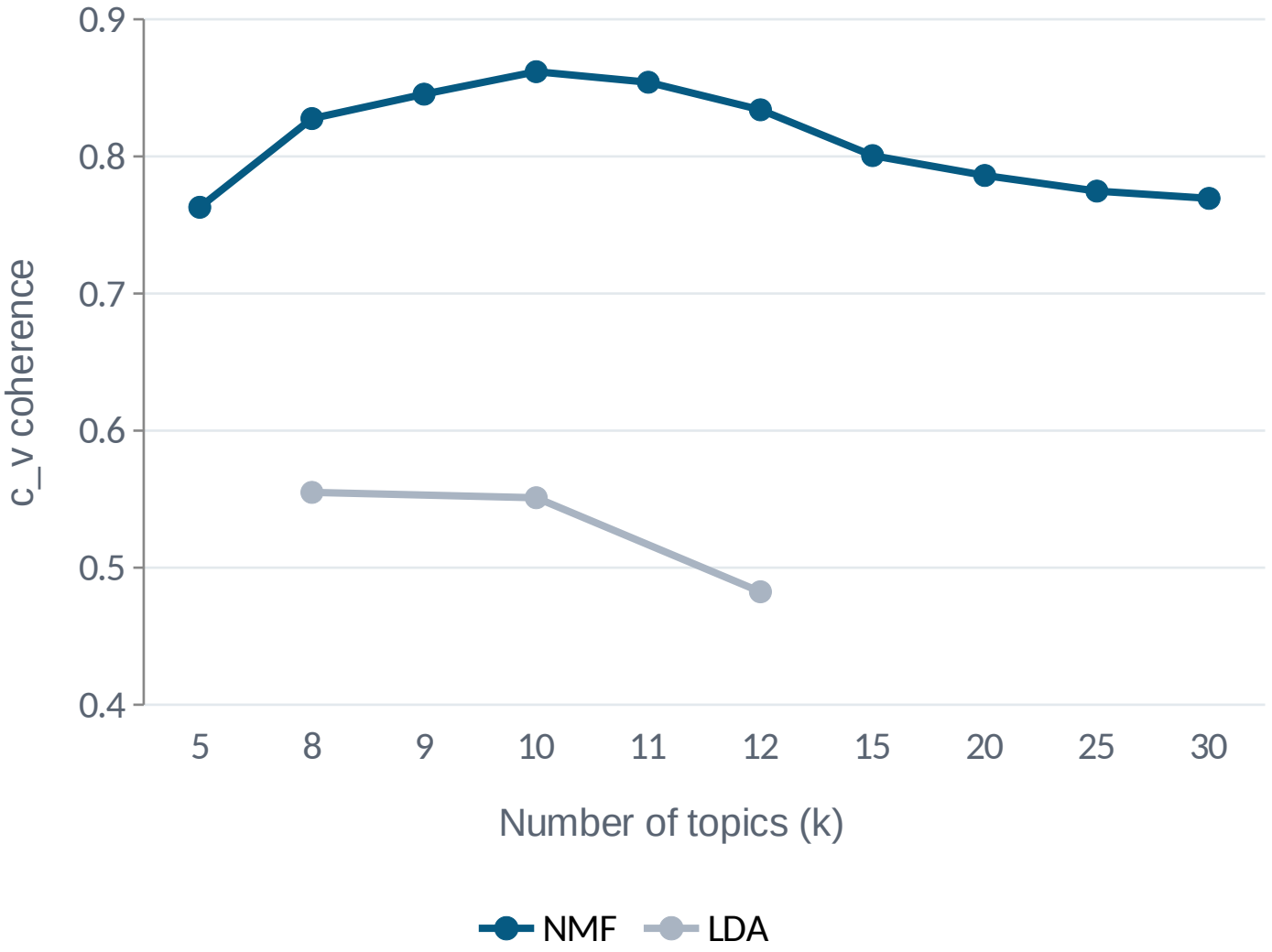
The distribution is bimodal with almost nothing in the middle: 5th percentile 0.071, 10th percentile 0.400.



Six times the threshold, under twice the effect.

Blind spot: it scores variety per account, so ten thousand accounts each posting one referral all look like people. That gap shows up in the topics.

Choosing k from evidence, not convention

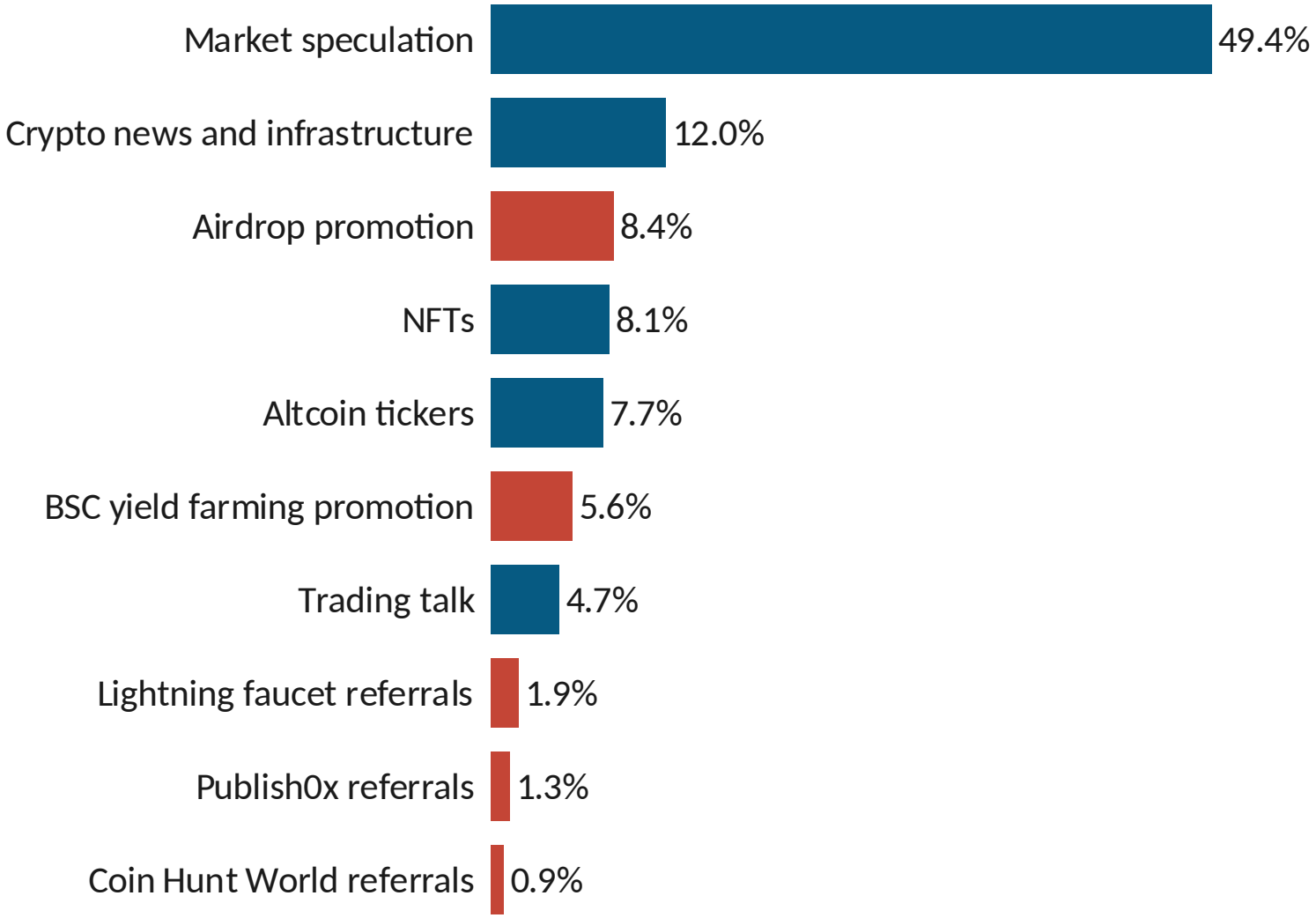


c_v coherence

- NMF peaks cleanly at k = 10 (0.8618)
- LDA is flat between 8 and 10, a gap of 0.004
- LDA never clears 0.56 at any k
- Tweets are too short for LDA's mixture assumption

Reconstruction error falls monotonically, 441.5 to 432.4, with no turning point. It would always say add more topics, which is why it cannot select k.

Ten topics, five of them not conversation



Conversation

Promotional template campaign

18.1%

of the sample sits in the five promotional topics.

One topic, market speculation, absorbs half the corpus as a residual: price guessing, plus everything that fit nowhere else.

Two sentiment methods, 42% agreement

VADER

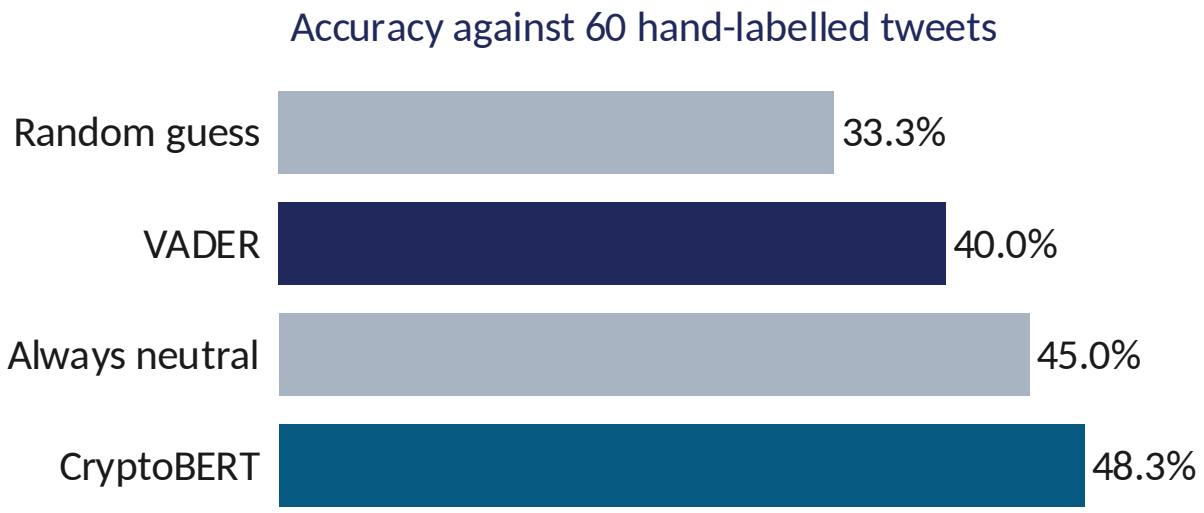
Rule-based lexicon, extended with 16 crypto terms we weighted by hand.

Bullish precision 0.26: three in four tweets it calls positive are not bullish. It reads cheerfulness as market optimism.

CryptoBERT

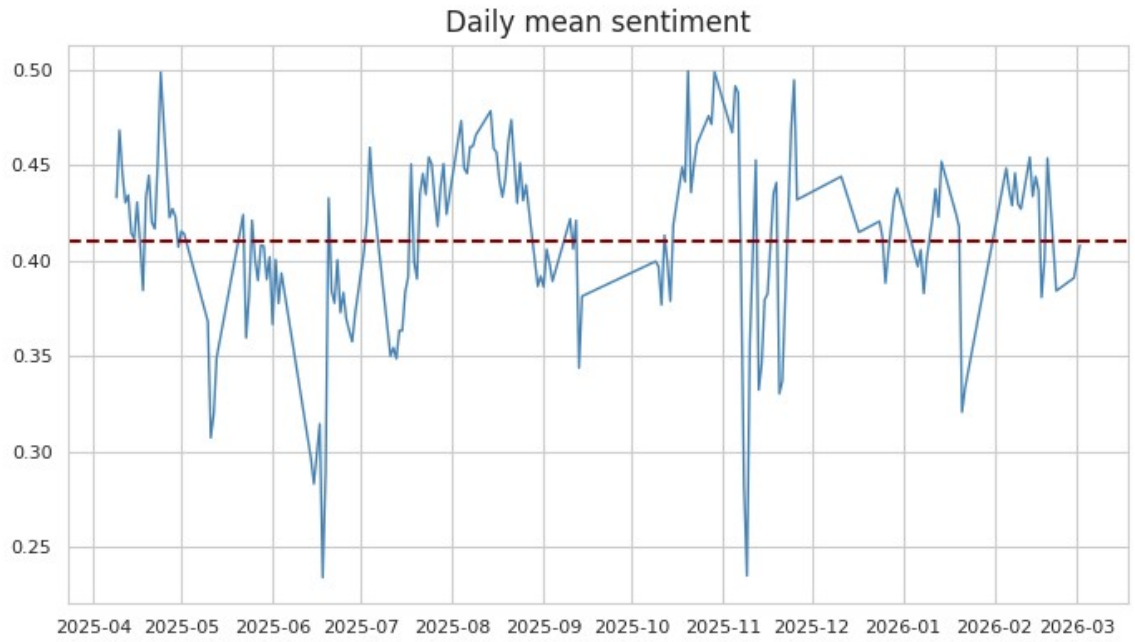
Transformer trained on crypto social media. Returns Bearish, Neutral, Bullish.

Bearish recall 0.72 against VADER's 0.44, which is the half of the signal volatility cares about.

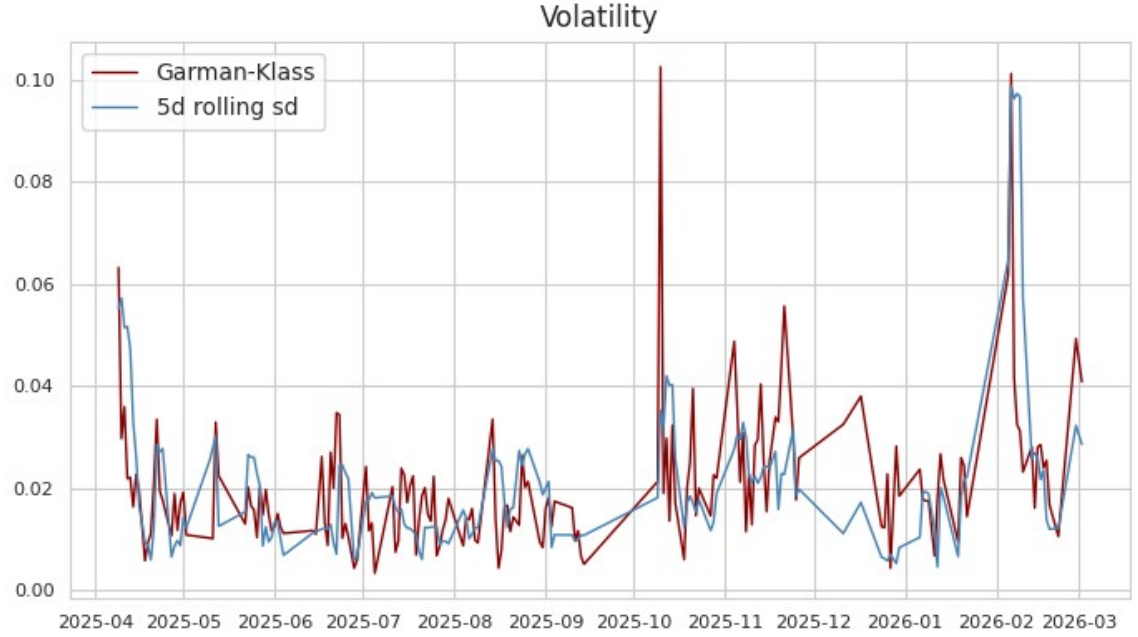


60 tweets labelled by hand, two raters, Cohen's kappa 0.686. Always guessing neutral scores 45%, so neither method is good. Both read promotional spam as bullish enthusiasm, which is the Question 1 template problem again.

Sentiment barely moves while volatility spikes



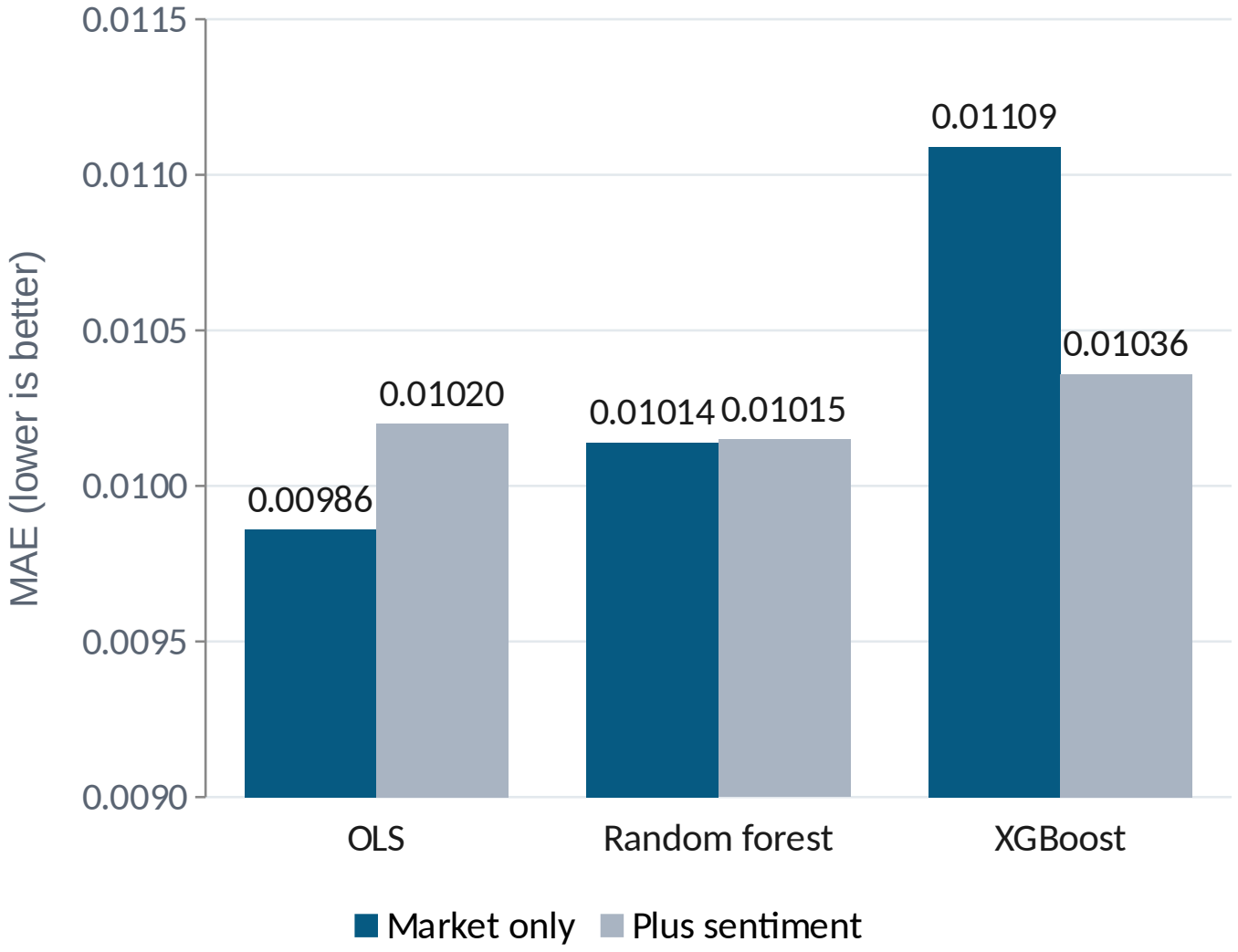
Daily mean CryptoBERT polarity, dashed line = period mean



Bitcoin volatility: Garman-Klass and 5-day rolling sd

Daily sentiment stays in a narrow, mostly positive band, with only about 6% of tweets labelled bearish. Volatility sits near 0.02 for most of the period and jumps sharply in October 2025 and February 2026. A series this flat has little variation to explain those spikes with.

Sentiment does not improve the forecast



- OLS gets worse, random forest unchanged
- XGBoost improves toward the OLS baseline, not past it, which reads as regularisation
- Directional accuracy: 36% to 52% across all six models
- Joint F-test on the sentiment block: p = 0.373

Expanding window, refit every step, 59 one-day-ahead forecasts. Differences this small are not distinguishable from noise.

The dataset's dates are out by four years

BERTopic surfaced Tokyo Olympics, covid vaccination and NFT mints. That is 2021. The date column says 2025 to 2026.

Probe	Dataset says	Actually happened
Tokyo Olympics	93% in July 2025	Opening ceremony 23 July 2021
Covid vaccination	94% in July 2025	Rollout mid-2021
Bitcoin Pizza Day	91% in May 2025	22 May, annually
El Salvador	Peak June 2025	Bitcoin Law announced June 2021
Beijing	Cluster February 2026	Winter Olympics February 2022

Question 1 is unaffected: topic models depend on the ordering of the calendar, not its absolute values. Question 2 pairs 2021 tweets with 2025 prices, so its results demonstrate the pipeline rather than establish anything about Bitcoin. The correction is one line.

What we would change

- Correct the four-year offset before the price merge. One line.
- Filter templates on the first eight tokens, not the whole string. That clears five topics and probably splits the residual.
- Weight daily sentiment by tweet count, so days with 296 tweets stop contributing the same noise as days with 49,036.

THANK YOU